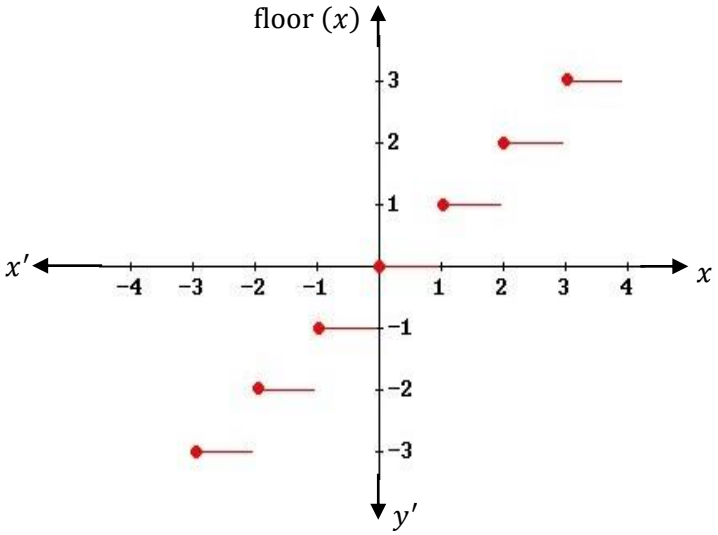
Chapter 3

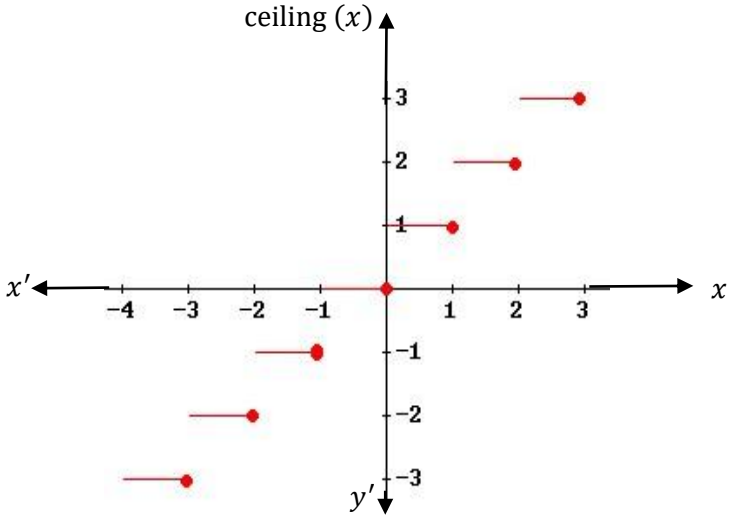
Functions

- **Lecture Contents:**
  - ✓ 1.1 (Page 10): Basic Concepts on Functions.
  - ✓ 1.1 (Page 15): The Vertical Line Test
  - ✓ 1.1 (Page 15): Piecewise functions.
  - ✓ 1.1 (Page 16): Absolute Value function.
  - ✓ 1.1 (Page 16): Example 7 & 8.
  - ✓ 1.1 (Page 17): Symmetry.
  - ✓ Also discuss on skew-symmetric function.
- **Floor and Ceiling Functions:**
  Let x be a real number.The floor function of x, denoted by $f(x) = \lfloor x \rfloor$, is the largest integer that is smaller than or equal to x.



Example: $\lfloor 0.7 \rfloor = 0, \lfloor 1.3 \rfloor = 1$.

The ceiling function of x, denoted by $f(x) = \lceil x \rceil$, is the smallest integer that is larger than or equal to x.

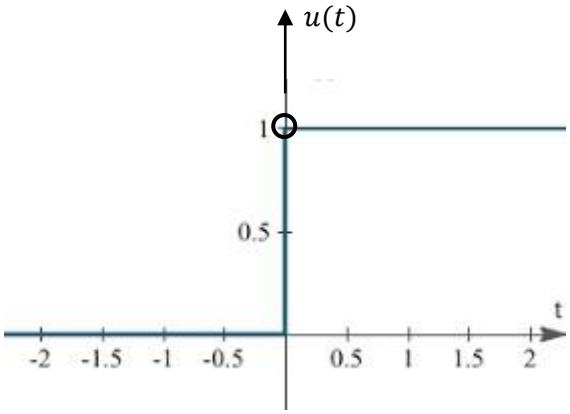


Example: $\lceil 0.3 \rceil = 1, \lceil 2.1 \rceil = 3$.

- **Unit Step Function:**
  The unit step function is defined as follows:

$$u(t) = \begin{cases} 0; \text{for } t < 0 \\ 1; \text{for } t > 0 \end{cases}$$

We do not define $u(t)$ at $t = 0$. Rather, at $t = 0$ we think of it as in transition between 0 and 1. It is called the unit step function because it takes a unit step at $t = 0$. It is sometimes called the Heaviside function.



Sample Exercise Set 3.1

1. Find domain and range of the following functions:
(a) $f(x) = \sqrt{x+2}$, (b) $f(x) = \frac{1}{x^2 - x}$, (c) $f(x) = x + \sqrt{2-x}$, (g) $f(x) = \lfloor x \rfloor$, (h) $f(x) = \lceil x \rceil$, and (i) $u(t-4)$.

2. Exercise 1.1: (Page 21) 41 and 43.
3. Exercise 1.1: (Page 22) 57, 61, 62 and 63.

- **Lecture Contents:**
  - ✓ 1.3 (Page 36-40): Transformation of Functions.

Sample Exercise Set 3.2

Sketch the graph of the following functions:

(a) $f(x) = x^2 + 2$, (b) $f(x) = -(x+2)^2 + 2$, (c) $f(x) = -x^3 + 2$,

(d) $f(x) = (x-1)^3$, (e) $f(x) = \ln(x+1)$, (f) $f(x) = -e^{x+3}$,

(g) $f(x) = \frac{1}{x+2}$, (h) $f(x) = \sin 2x$, (i) $f(x) = -\cos x$, (j) $f(x) = |x-2| + 1$,

(k) $f(x) = \sqrt{2-x} + 1$, (l) $f(x) = \sinh x$ and (m) $f(x) = \cosh x$.